Problem Set 9

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: ISL 4; CO 7.1, 9.3; MML 7.1. Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited exercise numbers are from ISL, 2nd edition (James et al., 2021).

1. ISL, Chapter 4, Conceptual exercises: problems 1, 6.
2. ISL, Chapter 4, Applied exercises: a logistic-regression exercise, e.g. on the `Weekly` data set **[code]**.
3. Derive the gradient of the cross-entropy (logistic) loss with respect to the weight vector for a single training example, and show that the resulting gradient-descent update has a simple "prediction minus label" form.
4. **[code]** Implement gradient descent to fit logistic regression on a small two-dimensional data set. Plot the loss versus iteration and the resulting decision boundary.
5. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).